23BEC0503

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF ELECTRONICS ENGINEERING

Fall Semester 2025-2026

BECE406E – FPGA Based System Design

TASK 4 – Static Timing Analysis

SANJAY ELAVARASAN KARTHIKEYAN

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Aim_1:

Display the time in the format Hr: Min: Sec using clock divider concept and display the output using seven segment display. Attach screenshots of code and FPGA board implementation.

Software Details:

- Functional Simulation: Modelsim
- Synthesis & Implementation: Intel Quartus Prime 23.1

Hardware Details:

- Family: Cyclone IV E
- Package: FBGA
- Pin count: 780
- Core speed grade – 7
- Device - EP4CE115F29C7

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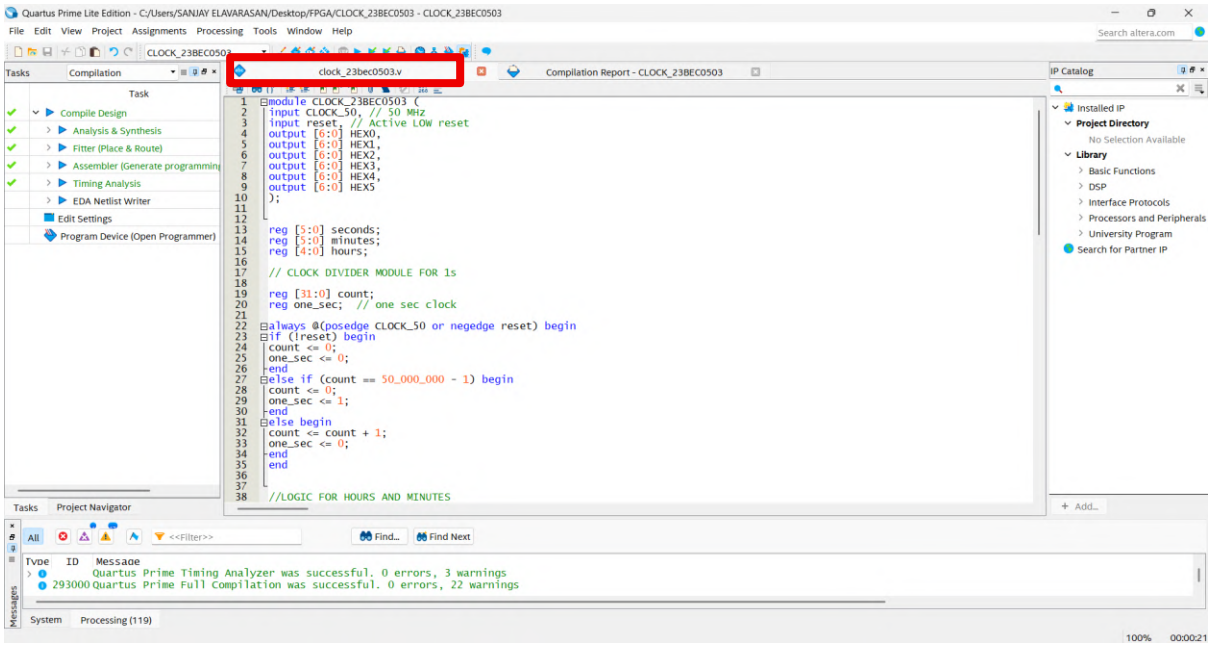
RTL Code for Clock:




```
//LOGIC FOR HOURS AND MINUTES
always @(posedge CLOCK_50 or negedge reset) begin
if (!reset) begin
seconds <= 0;
minutes <= 0;
hours <= 0;
end
else if (one_sec) begin

if (seconds == 59) begin
seconds <= 0;

if (minutes == 59) begin
minutes <= 0;

if (hours == 23)
hours <= 0;
else
hours <= hours + 1;

end
else
minutes <= minutes + 1;

end
else
seconds <= seconds + 1;

end
end
```

```
wire [3:0] sec_unit = seconds % 10;
wire [3:0] sec_tens = seconds / 10;

wire [3:0] min_unit = minutes % 10;
wire [3:0] min_tens = minutes / 10;

wire [3:0] hr_unit = hours % 10;
wire [3:0] hr_tens = hours / 10;


segment7 s0({1'b0,sec_unit}, HEX0);
segment7 s1({1'b0,sec_tens}, HEX1);

segment7 s2({1'b0,min_unit}, HEX2);
segment7 s3({1'b0,min_tens}, HEX3);

segment7 s4({1'b0,hr_unit}, HEX4);
segment7 s5({1'b0,hr_tens}, HEX5);

endmodule
```

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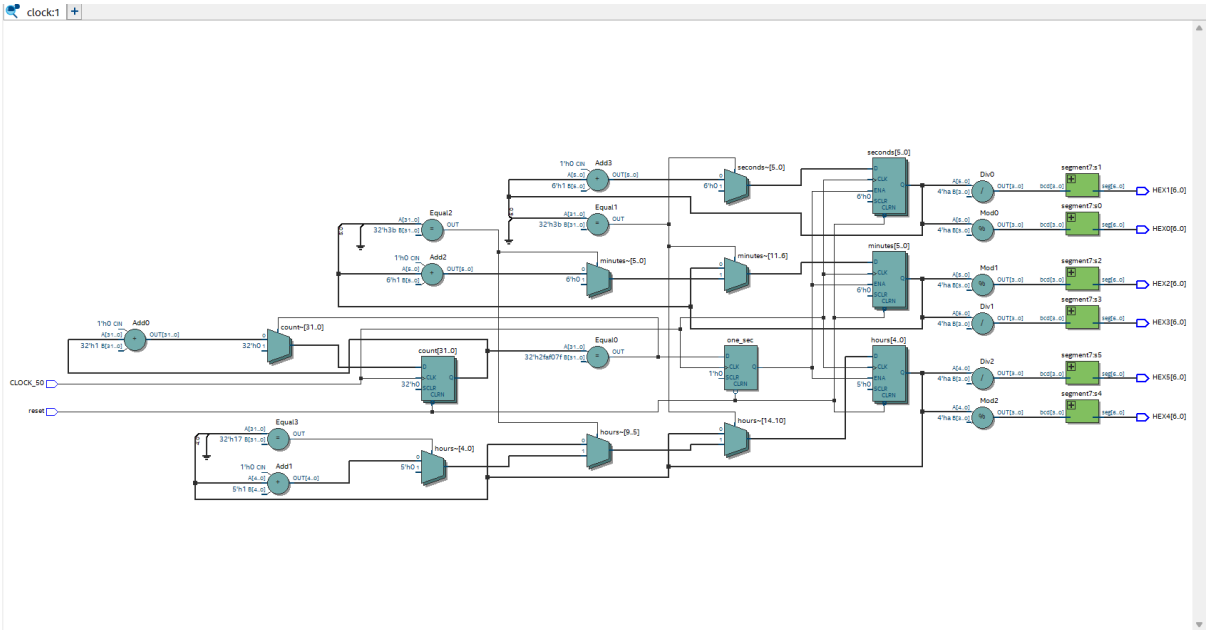
```
module segment7(
  input [3:0] bcd,
  output reg [6:0] seg
);
 always @(*) begin
 case (bcd)
  4'd0: seg = 7'b1000000;
  4'd1: seg = 7'b1111001;
  4'd2: seg = 7'b0100100;
  4'd3: seg = 7'b0110000;
  4'd4: seg = 7'b0011001;
  4'd5: seg = 7'b0010010;
  4'd6: seg = 7'b0000010;
  4'd7: seg = 7'b1111000;
  4'd8: seg = 7'b0000000;
  4'd9: seg = 7'b0010000;
  default: seg = 7'b1111111;
  endcase
  end
 endmodule
```

RTL VIEW :



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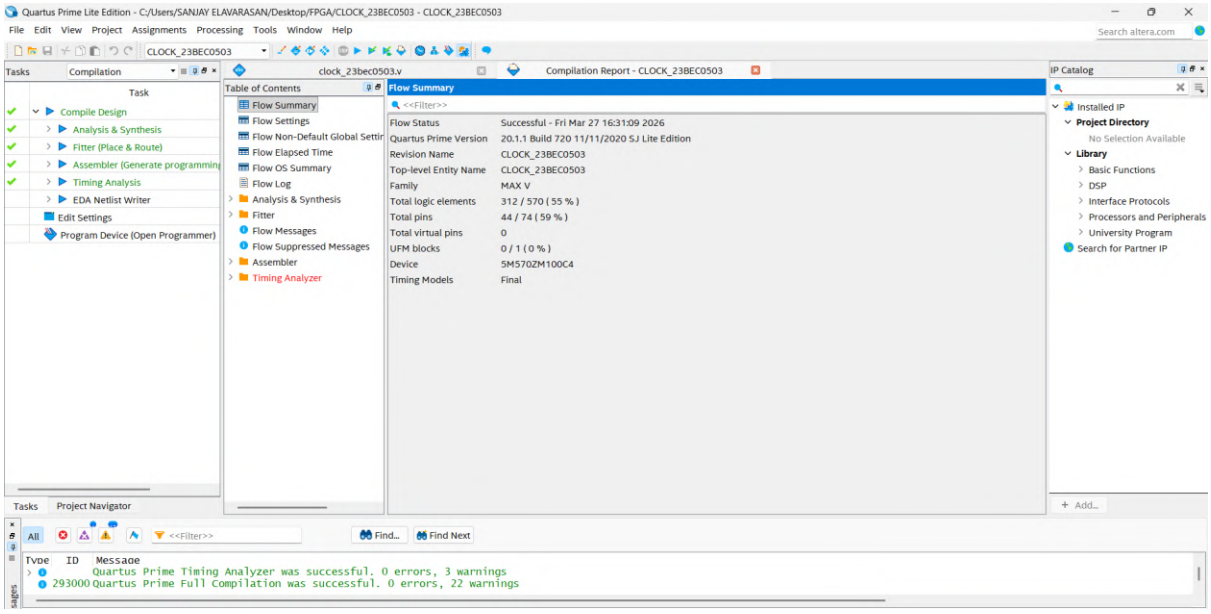
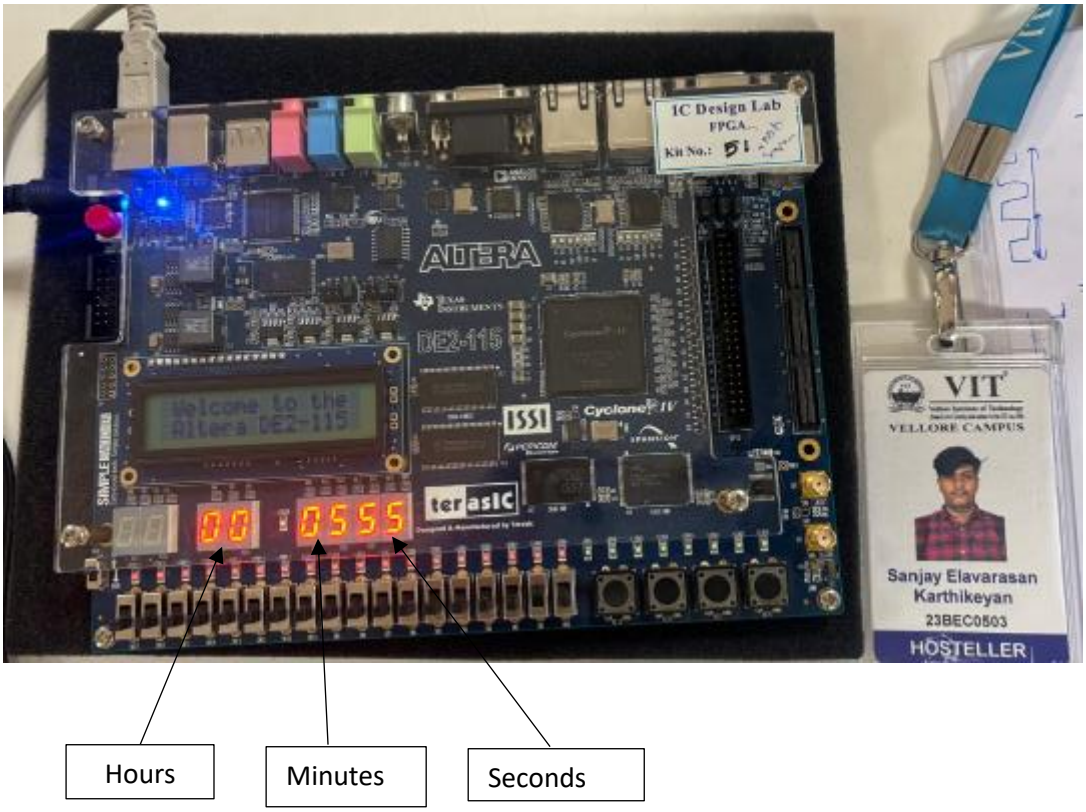
COMPILATION REPORT :



FPGA IMPLEMENTATION



Hours

Minutes

Seconds

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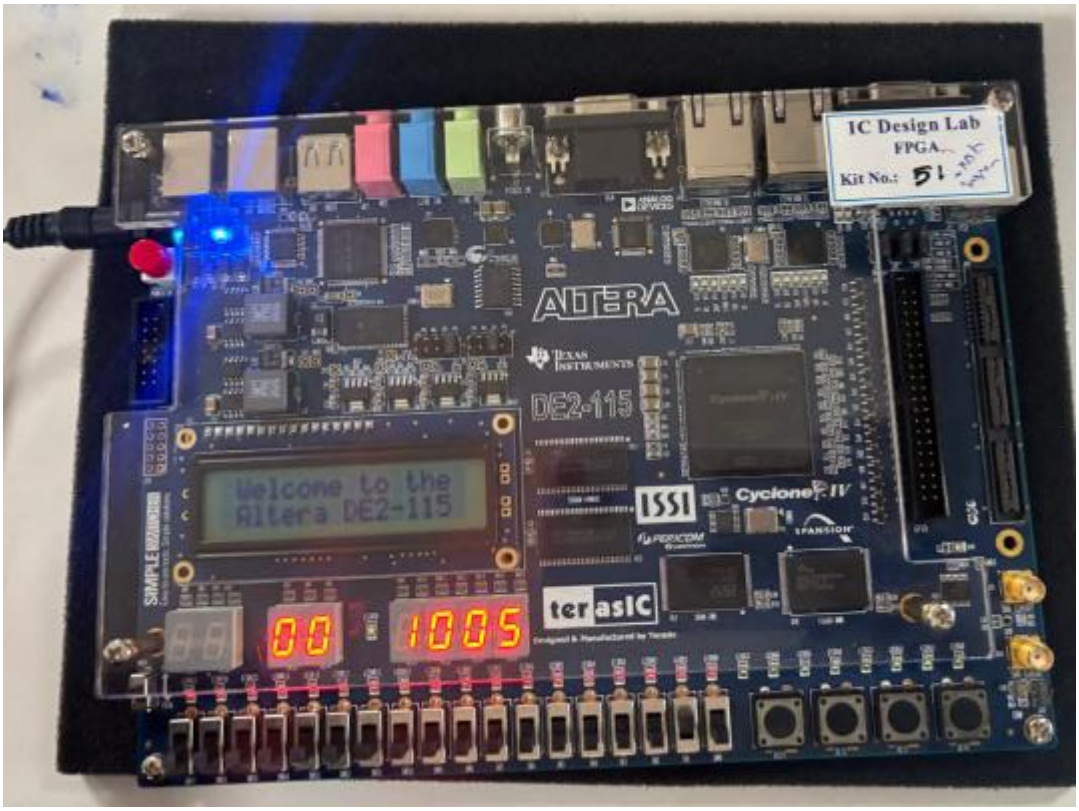
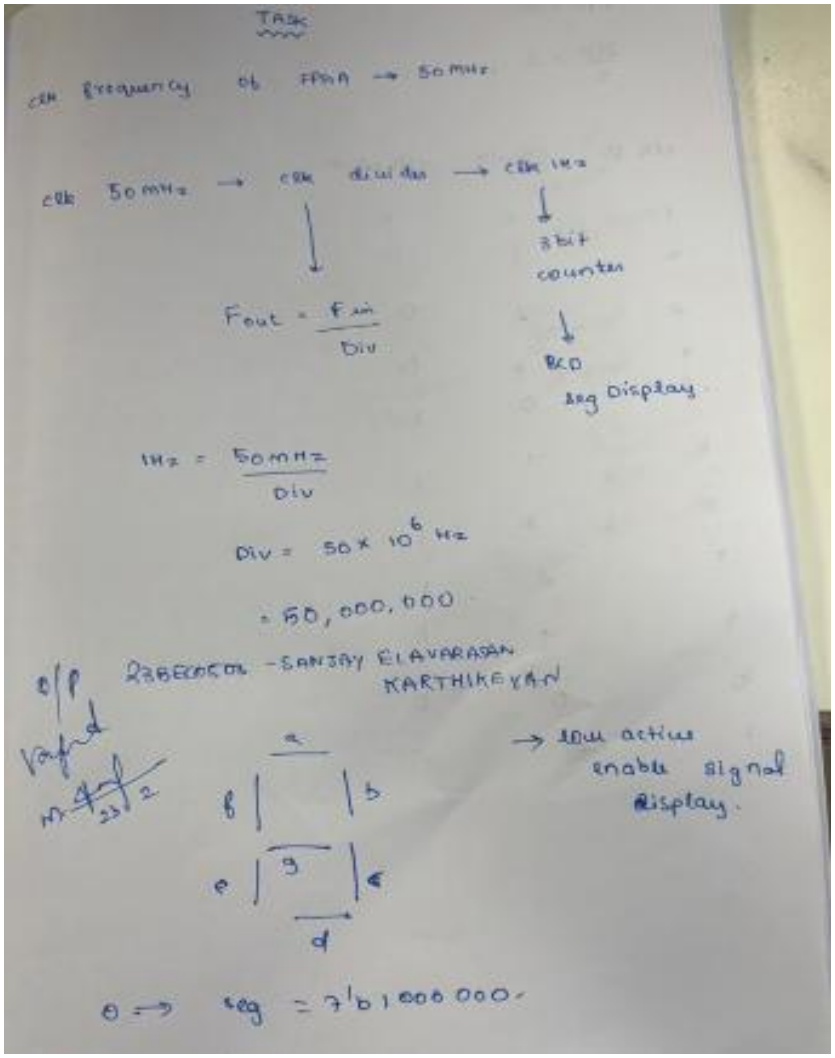


LAB ATTESTATION



TASK

clk frequency of FPGA → 50 MHz

clk 50 MHz → clk divider → clk 1 Hz

↓ 3 bit counter

↓ BCD seg Display

$F_{out} = \frac{F_{in}}{Div}$

$1Hz = \frac{50MHz}{Div}$

$Div = 50 \times 10^6 Hz$

$= 50,000,000$

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→ low active enable signal display.

0 → seg = 7'b1000000

Aim_2:

Implement Synchronous FIFO using Quartus and verify the functionality. Verify the timing analysis and power analysis. Check for the set and hold violation and find the critical path. Attach the RTL view of FIFO and Screenshots of code and outputs.

Software Details:

- Functional Simulation: Modelsim
- Synthesis & Implementation: Intel Quartus Prime 23.1

Hardware Details:

- Family: Cyclone IV E
- Package: FBGA
- Pin count: 780
- Core speed grade – 7
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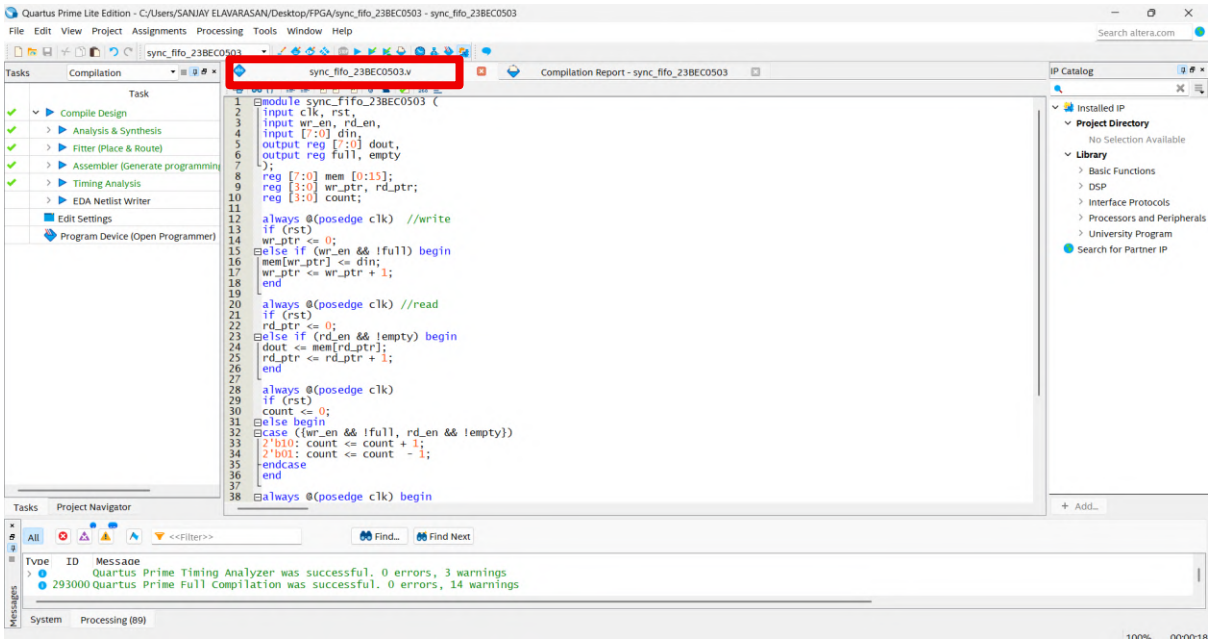
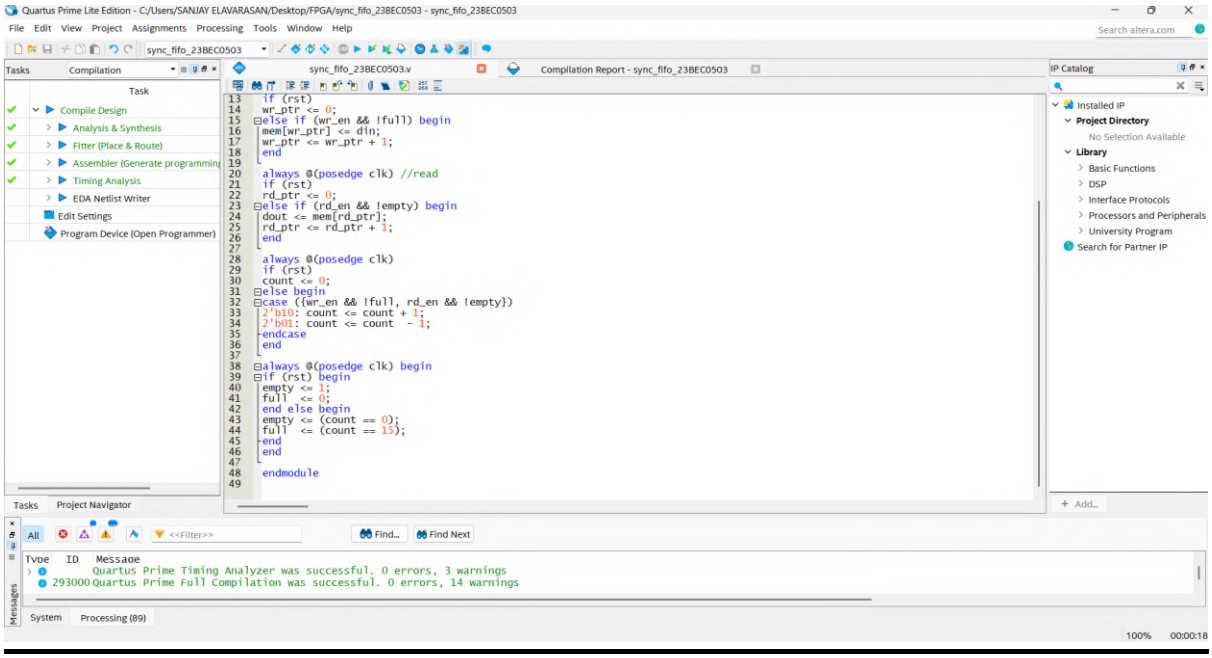
RTL Code for FIFO:





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TESTBENCH FOR FUNCTIONAL VERIFICATION

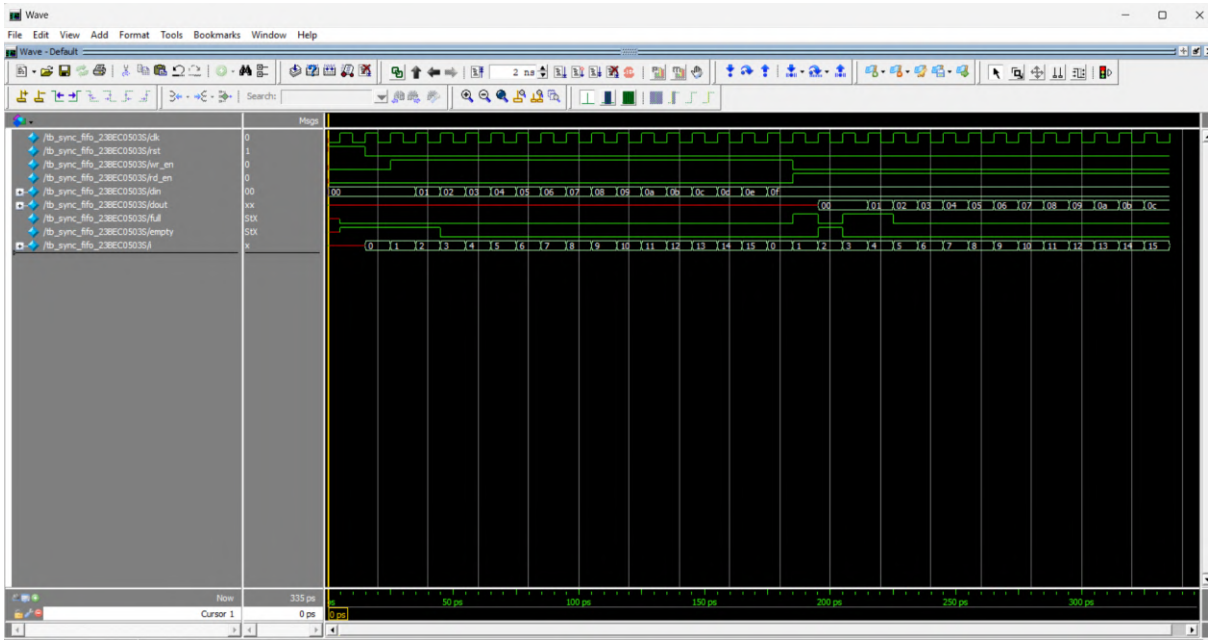
File Edit View Tools Bookmarks Window Help

C:/Users/SANJAY ELAVARASAN/Desktop/FPGA/DA_4/sync_fifo_tb.v - Default *

```
module tb_sync_fifo_23BEC0503S;

reg clk, rst;
reg wr_en, rd_en;
reg [7:0] din;
wire [7:0] dout;
wire full, empty;

// Instantiate DUT
sync_fifo dut (
    .clk(clk),
    .rst(rst),
    .wr_en(wr_en),
    .rd_en(rd_en),
    .din(din),
    .dout(dout),
    .full(full),
    .empty(empty)
);

////////////////////////////////////////////////////
// Clock generation (only place using # delay)
////////////////////////////////////////////////////
always #5 clk = ~clk;

////////////////////////////////////////////////////
// Test sequence
////////////////////////////////////////////////////
integer i;

initial begin
    $display("Starting FIFO Testbench...");
    clk = 0;
    rst = 1;
    wr_en = 0;
    rd_en = 0;
    din = 0;

    // Apply reset
    repeat(2) @(posedge clk);
    rst = 0;

    ////////////////////////////////////////////////
    // 1. WRITE UNTIL FULL
    ////////////////////////////////////////////////
    $display("\n--- Writing Data ---");

    for (i = 0; i < 16; i = i + 1) begin
        @(posedge clk);
        wr_en = 1;
        rd_en = 0;
        din = i;
    end


    ////////////////////////////////////////////////
    // 2. READ UNTIL EMPTY
    ////////////////////////////////////////////////
    $display("\n--- Reading Data ---");

    for (i = 0; i < 16; i = i + 1) begin
        @(posedge clk);
        wr_en = 0;
        rd_en = 1;
    end



    ////////////////////////////////////////////////
    // END SIMULATION
    ////////////////////////////////////////////////
    $display("\nTest Completed.");
    $finish;
end

////////////////////////////////////////////////////
// MONITOR (Event-based, no # delays)
////////////////////////////////////////////////////
always @(posedge clk) begin
    $display("Time=%0t | wr=%b rd=%b | din=%0d dout=%0d | count=%0d | full=%b empty=%b",
             $time, wr_en, rd_en, din, dout, dut.count, full, empty);
end

endmodule
```

23BEC0503

WAVEFORM

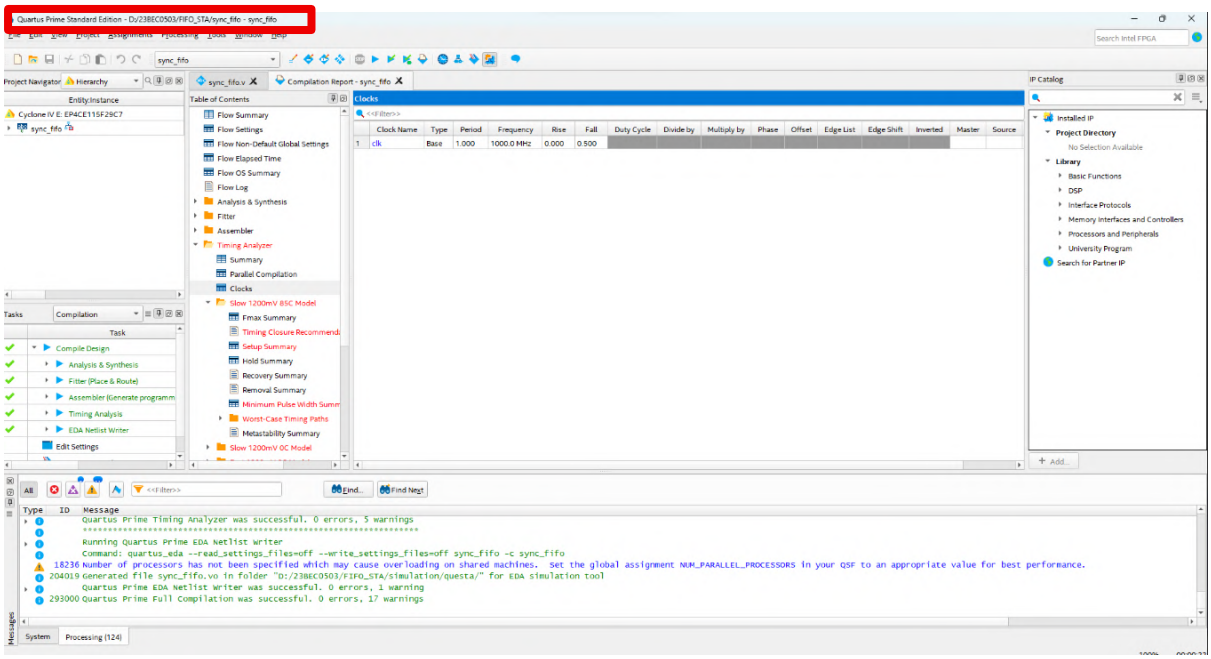


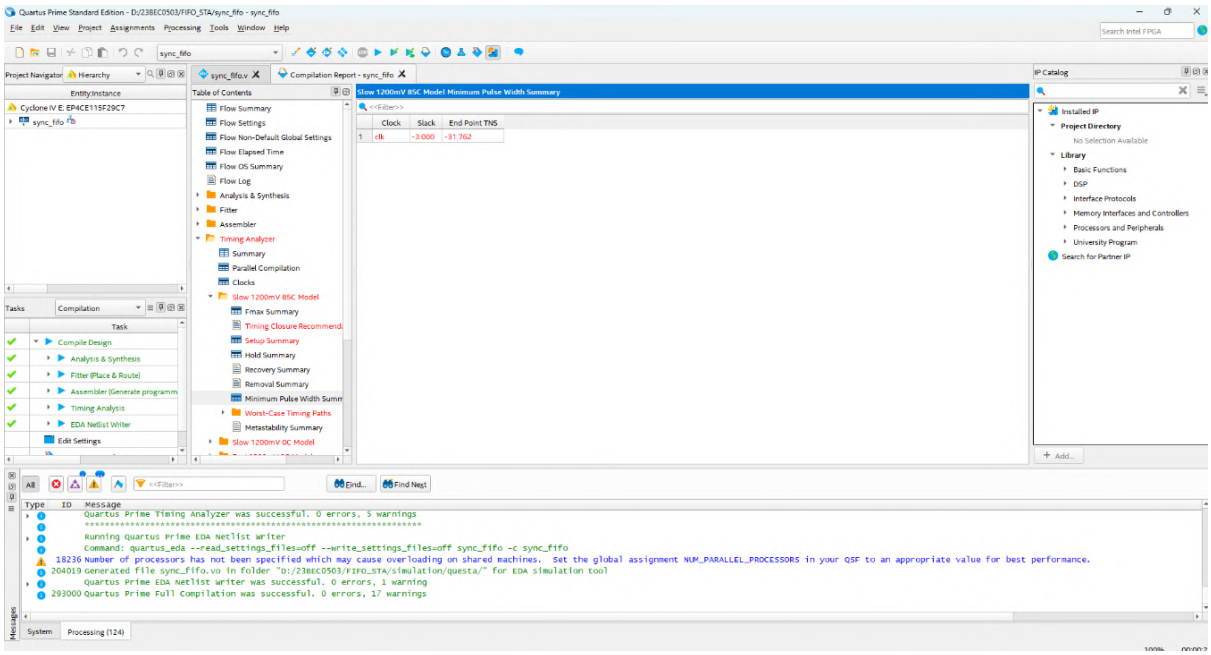

```
VSIM 3> run
# Starting FIFO Testbench...
# Time=5 | wr=0 rd=0 | din=0 dout=x | count=x | full=x empty=x
# Time=15 | wr=0 rd=0 | din=0 dout=x | count=0 | full=0 empty=1
#
# --- Writing Data ---
# Time=25 | wr=0 rd=0 | din=0 dout=x | count=0 | full=0 empty=1
# Time=35 | wr=1 rd=0 | din=0 dout=x | count=0 | full=0 empty=1
# Time=45 | wr=1 rd=0 | din=1 dout=x | count=1 | full=0 empty=1
# Time=55 | wr=1 rd=0 | din=2 dout=x | count=2 | full=0 empty=0
# Time=65 | wr=1 rd=0 | din=3 dout=x | count=3 | full=0 empty=0
# Time=75 | wr=1 rd=0 | din=4 dout=x | count=4 | full=0 empty=0
# Time=85 | wr=1 rd=0 | din=5 dout=x | count=5 | full=0 empty=0
# Time=95 | wr=1 rd=0 | din=6 dout=x | count=6 | full=0 empty=0
# Time=105 | wr=1 rd=0 | din=7 dout=x | count=7 | full=0 empty=0
# Time=115 | wr=1 rd=0 | din=8 dout=x | count=8 | full=0 empty=0
# Time=125 | wr=1 rd=0 | din=9 dout=x | count=9 | full=0 empty=0
# Time=135 | wr=1 rd=0 | din=10 dout=x | count=10 | full=0 empty=0
# Time=145 | wr=1 rd=0 | din=11 dout=x | count=11 | full=0 empty=0
# Time=155 | wr=1 rd=0 | din=12 dout=x | count=12 | full=0 empty=0
# Time=165 | wr=1 rd=0 | din=13 dout=x | count=13 | full=0 empty=0
# Time=175 | wr=1 rd=0 | din=14 dout=x | count=14 | full=0 empty=0
#
# --- Reading Data ---
# Time=185 | wr=1 rd=0 | din=15 dout=x | count=15 | full=0 empty=0
# Time=195 | wr=0 rd=1 | din=15 dout=x | count=0 | full=1 empty=0
# Time=205 | wr=0 rd=1 | din=15 dout=0 | count=15 | full=0 empty=1
# Time=215 | wr=0 rd=1 | din=15 dout=0 | count=15 | full=1 empty=0
# Time=225 | wr=0 rd=1 | din=15 dout=1 | count=14 | full=1 empty=0
# Time=235 | wr=0 rd=1 | din=15 dout=2 | count=13 | full=0 empty=0
# Time=245 | wr=0 rd=1 | din=15 dout=3 | count=12 | full=0 empty=0
# Time=255 | wr=0 rd=1 | din=15 dout=4 | count=11 | full=0 empty=0
# Time=265 | wr=0 rd=1 | din=15 dout=5 | count=10 | full=0 empty=0
# Time=275 | wr=0 rd=1 | din=15 dout=6 | count=9 | full=0 empty=0
# Time=285 | wr=0 rd=1 | din=15 dout=7 | count=8 | full=0 empty=0
# Time=295 | wr=0 rd=1 | din=15 dout=8 | count=7 | full=0 empty=0
# Time=305 | wr=0 rd=1 | din=15 dout=9 | count=6 | full=0 empty=0
# Time=315 | wr=0 rd=1 | din=15 dout=10 | count=5 | full=0 empty=0
# Time=325 | wr=0 rd=1 | din=15 dout=11 | count=4 | full=0 empty=0
# Time=335 | wr=0 rd=1 | din=15 dout=12 | count=3 | full=0 empty=0
#
# Test Completed.
# ** Note: $finish    : C:/Users/SANJAY ELAVARASAN/Desktop/FPGA/DA_4/sync_fifo_tb.v(73)
#    Time: 335 ps  Iteration: 1  Instance: /tb_sync_fifo_23BEC0503S
# 1
```

23BEC0503

TIMING AND POWER ANALYSIS

Before Timing Analysis:





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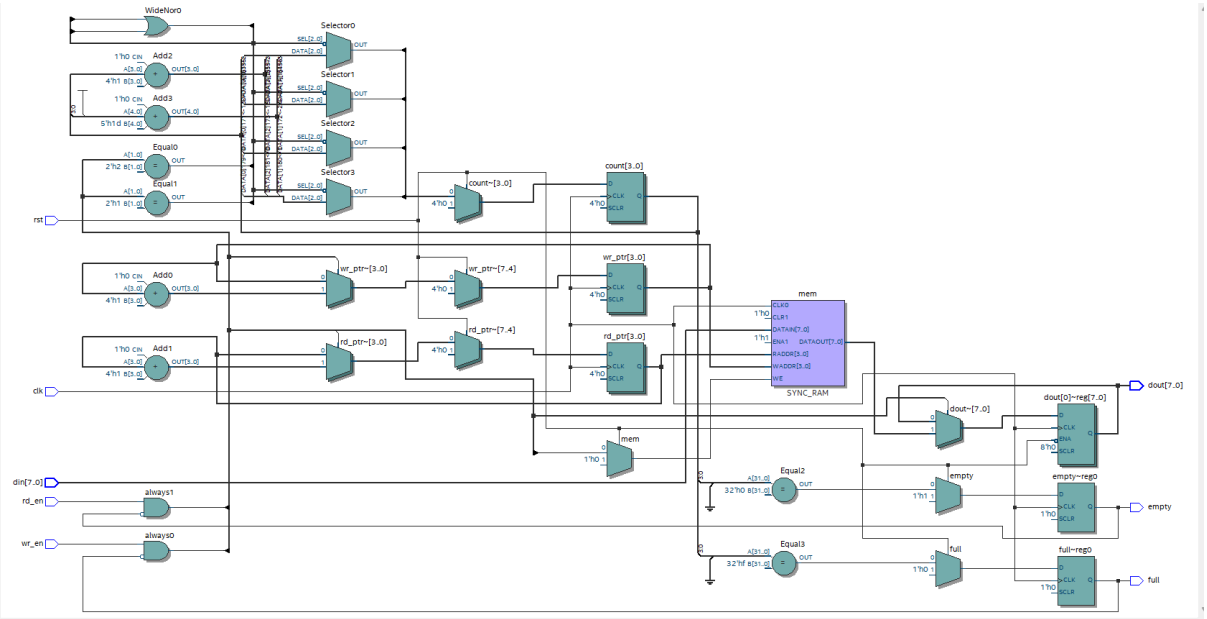
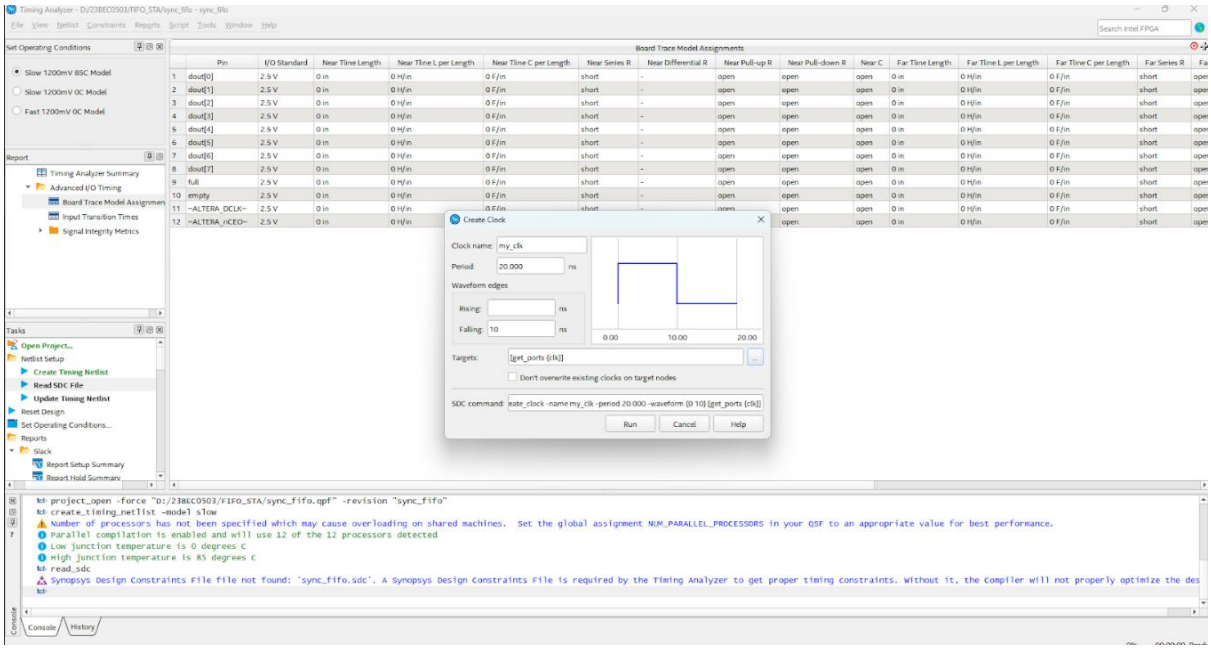
RTL VIEW



Creating Constraint File:

Clock :



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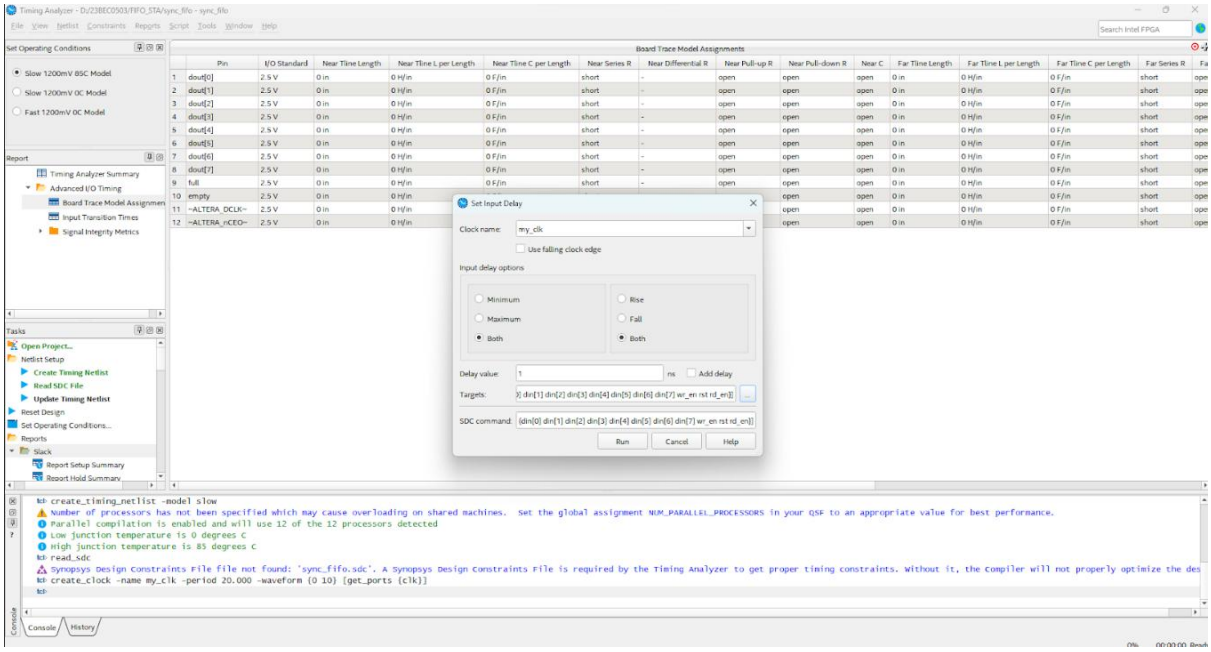
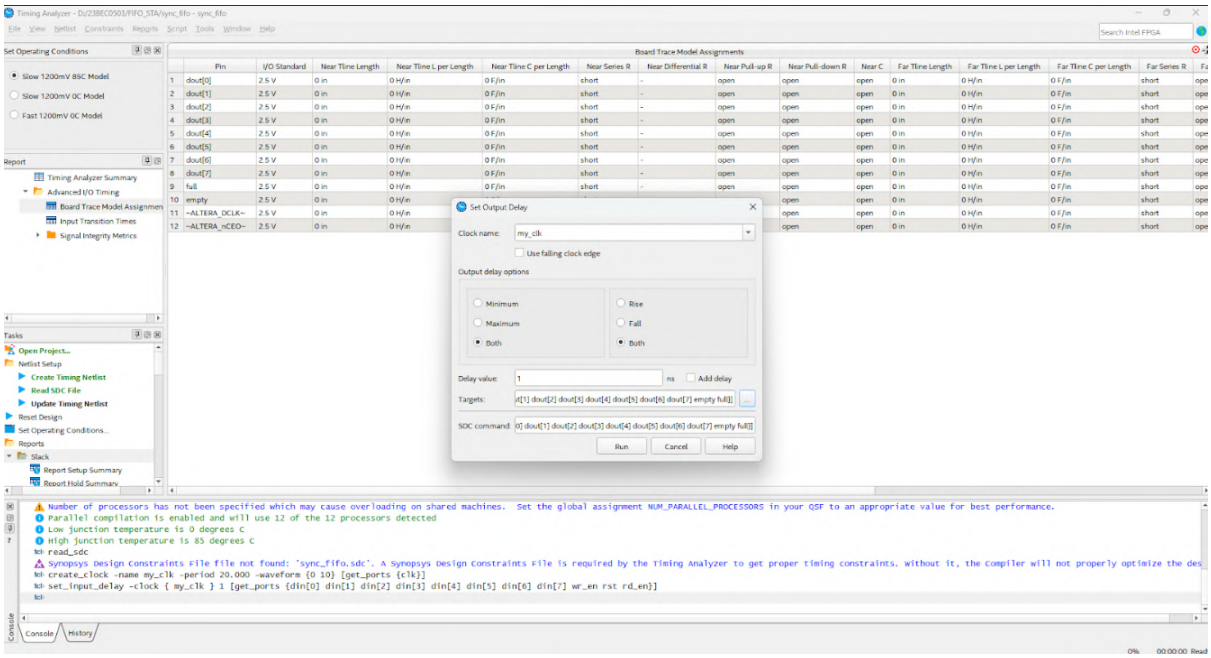
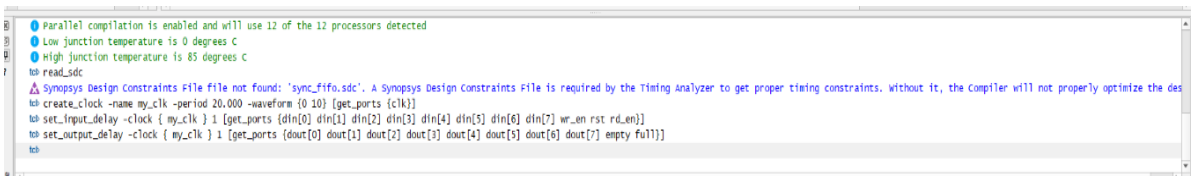
Input Delay :



Output Delay :



Final Commands :



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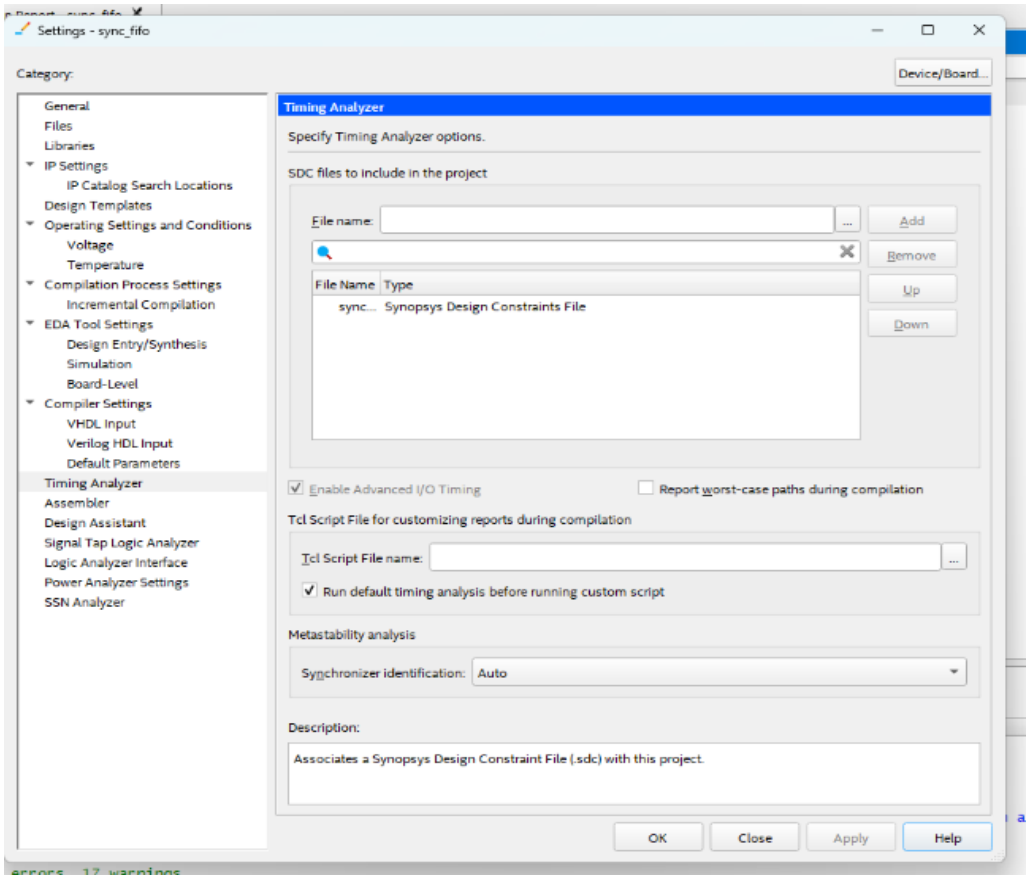
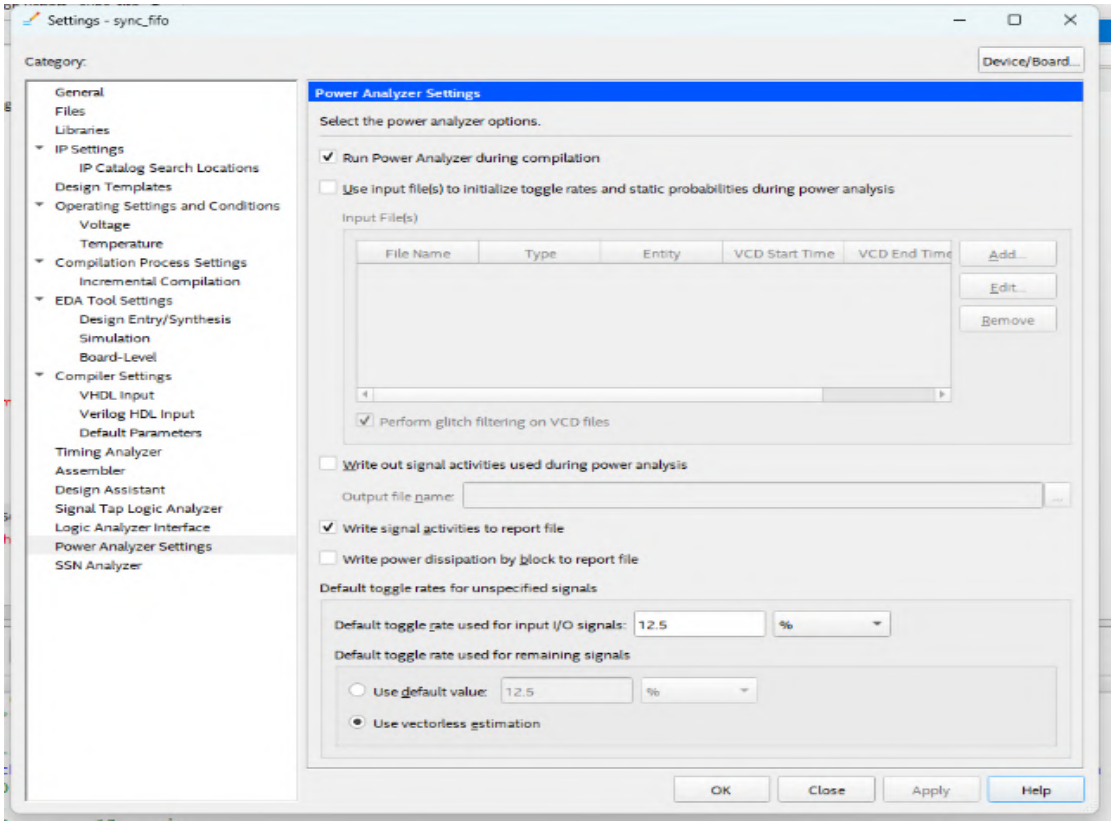
Timing Analyzer :



Power Analyzer :



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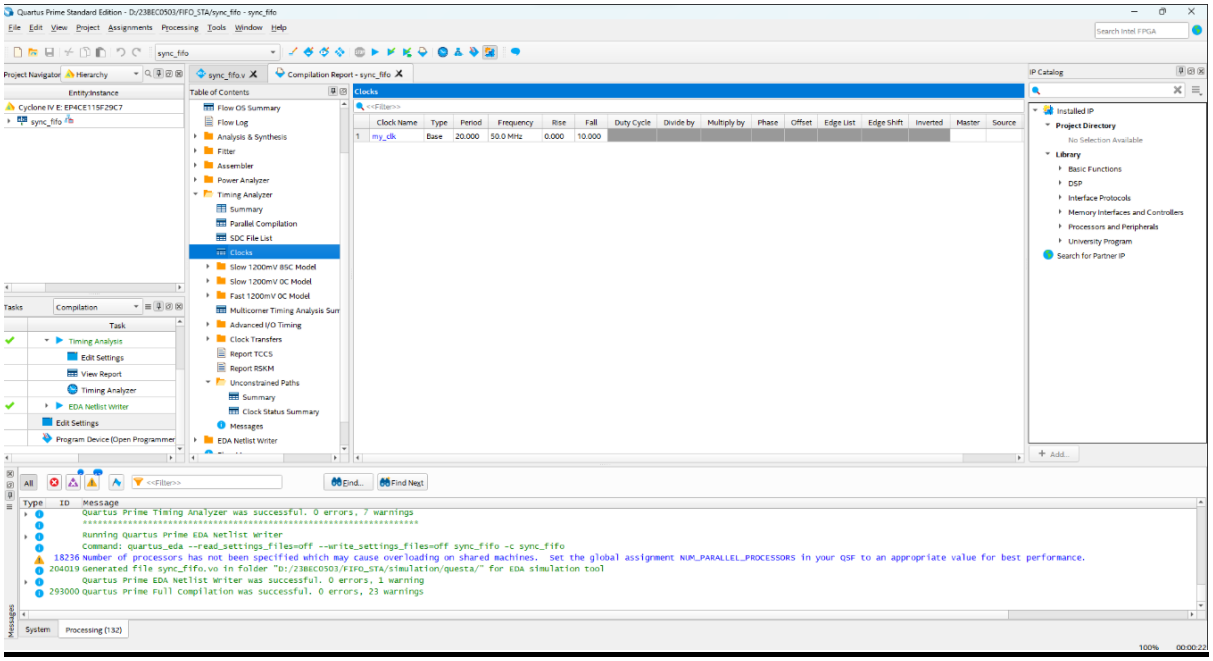
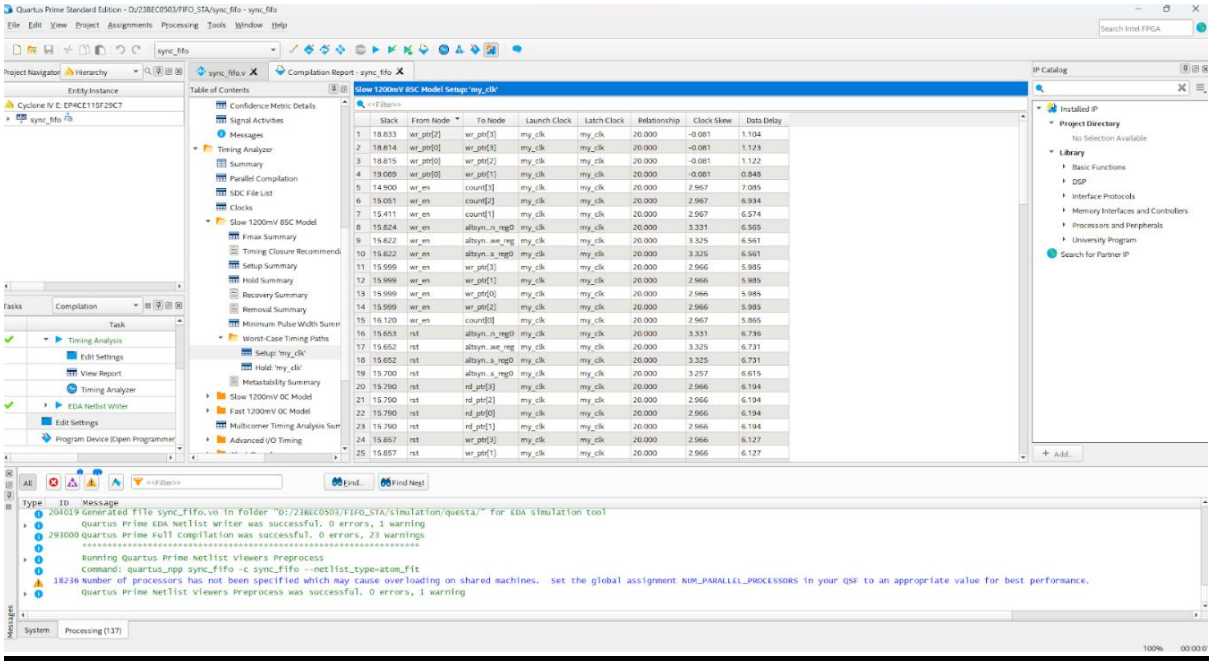
RESULTS AFTER STA:

TIMING RESULTS :



Setup :



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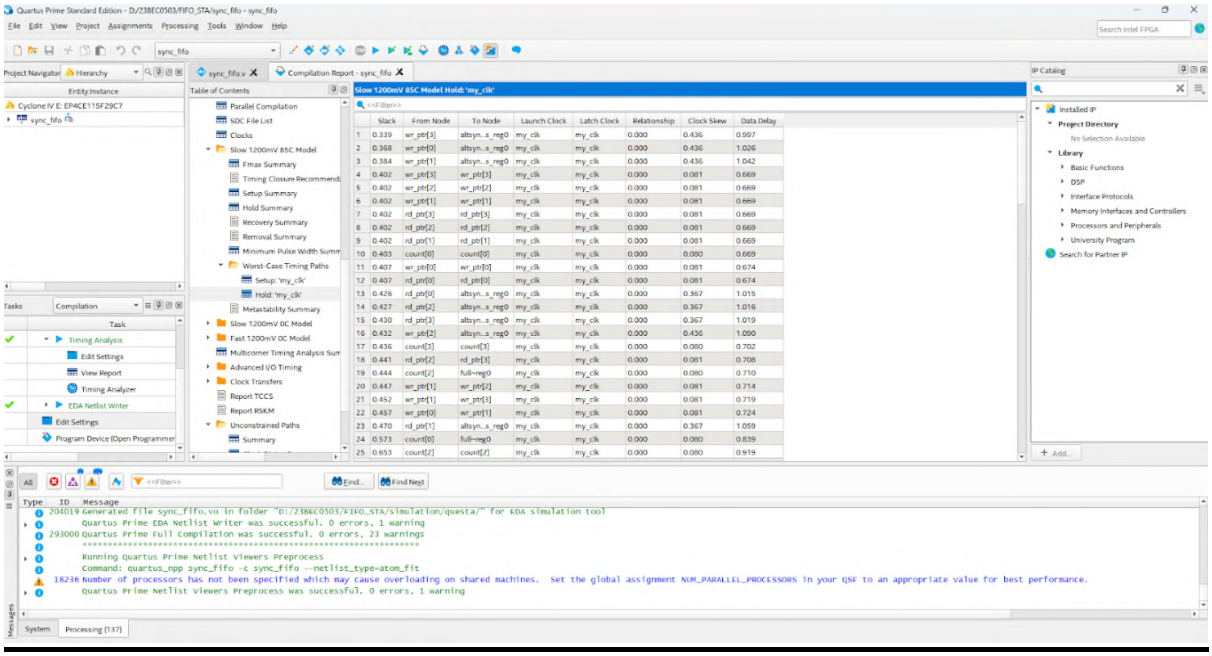
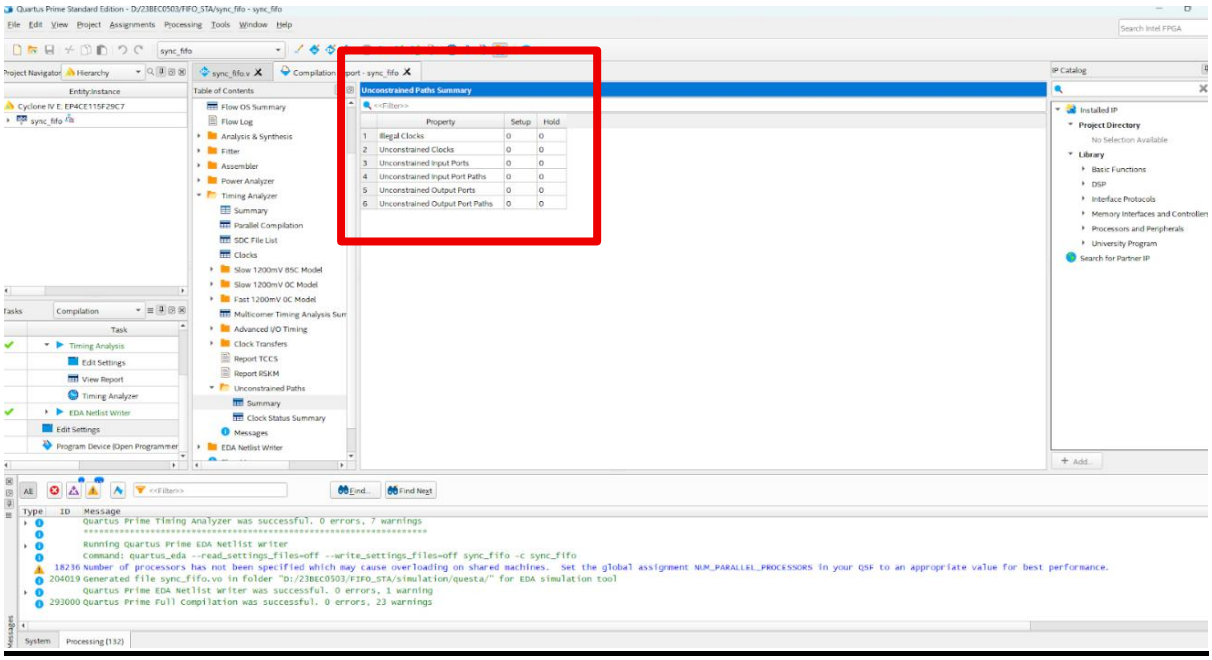
Hold:



Timing Analysis Summary:



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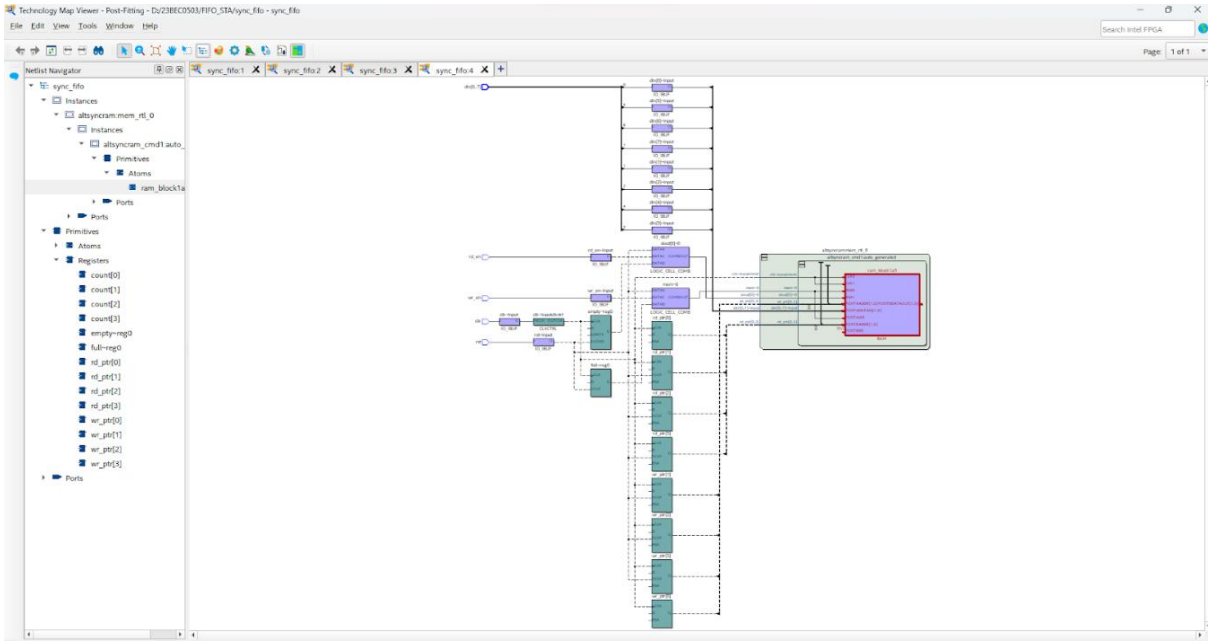
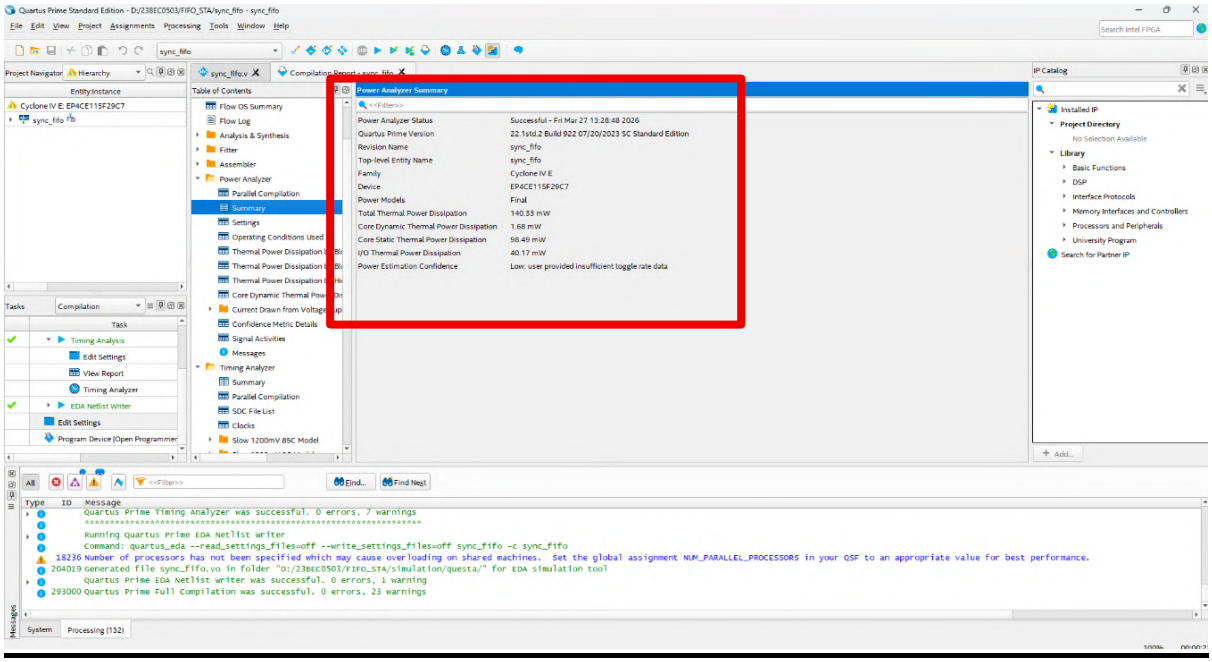
CRITICAL PATH :



POWER RESULTS :



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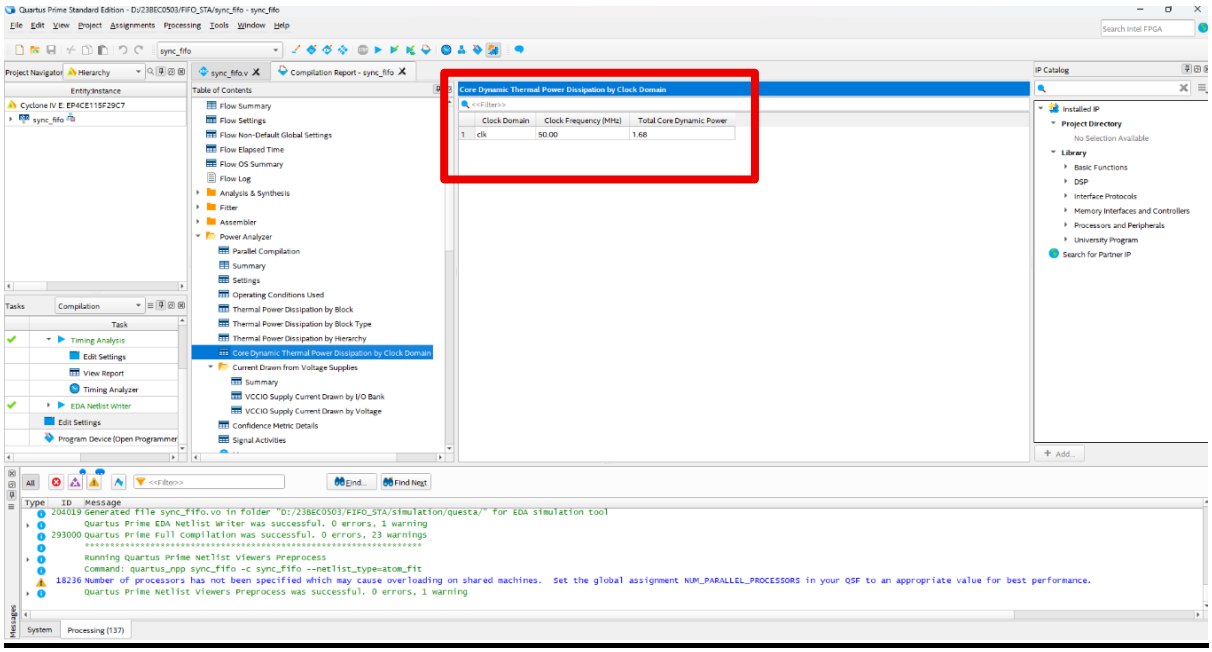

INFERENCE :

The experiment successfully demonstrated the design and implementation of a digital clock and a synchronous FIFO using Verilog in Quartus.

For the **digital clock**, the clock divider technique was effectively used to convert the 50 MHz input clock into a 1-second pulse. The time was accurately displayed in the format of hours, minutes, and seconds using seven-segment displays, confirming correct functionality of counting and rollover conditions (59→00 and 23→00).

For the **synchronous FIFO**, the design correctly handled data storage and retrieval operations based on write and read enable signals. The functionality was verified through simulation, ensuring proper operation of full and empty conditions. Timing analysis showed no setup and hold violations, indicating a stable design. The critical path was identified, and power analysis confirmed efficient resource utilization.

Overall, the experiment validated both functional correctness and timing reliability of the designs, demonstrating successful FPGA-based implementation using Quartus.